

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
```

```
df = pd.read_csv('/content/Mall_Customers.csv')
```

```
print("Dataset shape:", df.shape)
print(df.head())
```

```
Dataset shape: (200, 5)
  CustomerID  Genre  Age  Annual Income (k$)  Spending Score (1-100)
0           1   Male   19                15                39
1           2   Male   21                15                81
2           3  Female  20                16                 6
3           4  Female  23                16                77
4           5  Female  31                17                40
```

```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
X.head()
```

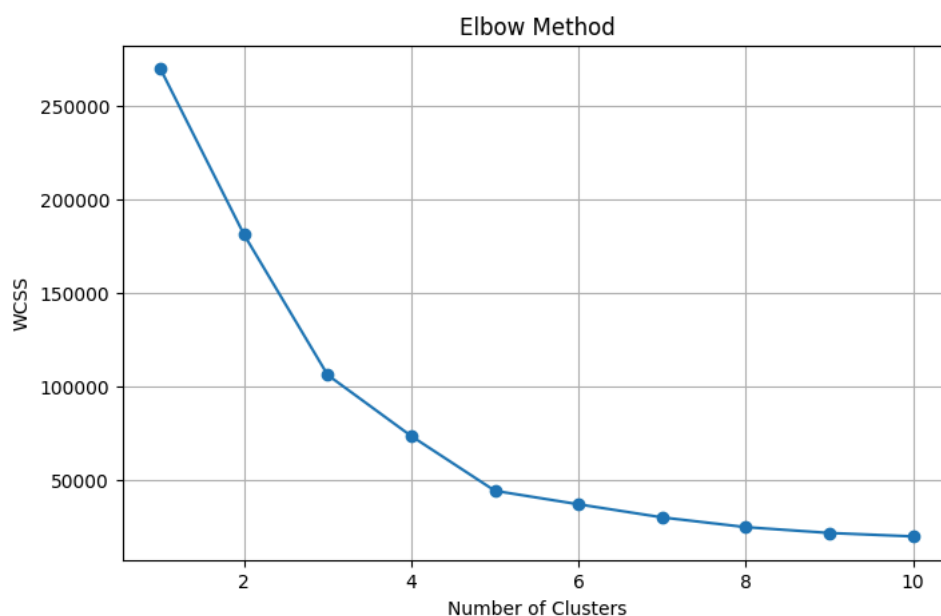
	Annual Income (k\$)	Spending Score (1-100)
0	15	39
1	15	81
2	16	6
3	16	77
4	17	40

Next steps: [Generate code with X](#) [New interactive sheet](#)

```
wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42, n_init=10)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

plt.figure(figsize=(8,5))
plt.plot(range(1,11), wcss, marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.grid(True)
plt.show()
```



```
kmeans = KMeans(n_clusters=5, random_state=42, n_init=10)

df['Cluster'] = kmeans.fit_predict(X)

print(df.head())
```

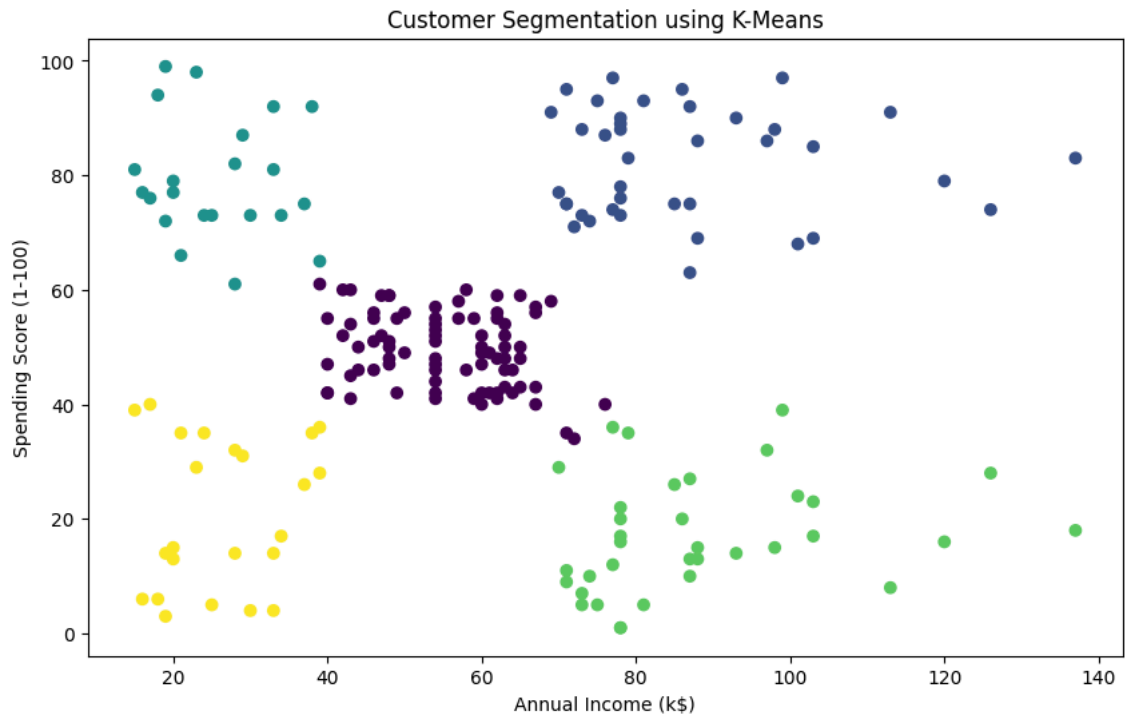
	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)	Cluster
0	1	Male	19	15	39	4
1	2	Male	21	15	81	2
2	3	Female	20	16	6	4
3	4	Female	23	16	77	2
4	5	Female	31	17	40	4

```
plt.figure(figsize=(10,6))

plt.scatter(
    df['Annual Income (k$)'],
    df['Spending Score (1-100)'],
    c=df['Cluster']
)

plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.title('Customer Segmentation using K-Means')

plt.show()
```



```
cluster_summary = df.groupby('Cluster')[['Annual Income (k$)', 'Spending Score (1-100)']].mean()

print(cluster_summary)
```

Cluster	Annual Income (k\$)	Spending Score (1-100)
0	55.296296	49.518519
1	86.538462	82.128205
2	25.727273	79.363636
3	88.200000	17.114286
4	26.304348	20.913043

Start coding or [generate](#) with AI.

